MIL-E-3118C

12 February 1974
SUPERSEDING
MIL-E-3118B
27 March 1967

MILITARY SPECIFICATION

ETCHERS, ELECTRIC, PORTABLE, 120-VOLT A.C.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric etchers of electric arc and vibrating point type.

1.2 Classification. Electric etchers furnished under this specification shall be of the following types, as specified (see 6.2):

Type I - Light duty etcher, 500 watts
Type II - Heavy duty etcher, 1,000 watts
Type III - Heavy duty etcher, vibrating type

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (For the Treatment of Communication, Electronic, and Associated Electrical Equipment)

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 5130

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Bureau of Standards

Handbook H-28 - Screw - Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Underwriters' Laboratories, Inc., Standards for Safety

UL 45 - Electric Tools
UL 115 - Asbestos and Asbestos Varnished Cloth Insulated Wires
UL 498 - Attachment Plugs and Receptacles
UL 499 - Electric Heating Appliances
UL 506 - Specialty Transformers

(Application for copies should be addressed to the Underwriters' Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingston Road, Northbrook IL 60062; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Standard product. The etchers delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.2 Codes and standards. The etchers shall comply with the applicable requirements of UL 45, 115, 498, 499, and 506.

3.2.1 Compliance. Prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative, satisfactory evidence that the etchers that he proposes to furnish under this specification meet the applicable requirements of UL 45, 115, 498, 499, and 506.

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3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating the etchers have been tested and conform to UL Standard No. 45, 115, 498, 499, and 506, as applicable.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial etcher, provided that the completed item complies with all provisions of this specification.

3.4 Design and construction. The etcher shall be designed to etch on metals by the electric-arc process for type I and II and vibratory-scarifying process for type III. Type I and II etchers shall be similar to figures 1 and 2 and type III similar to figure 3.

3.4.1 Electrical characteristics. The etchers shall operate on 120 volts (rated voltage), 50 to 60 Hertz, AC (alternating current).

3.4.2 Insulation resistance. Insulation of the etcher shall have a dielectric strength capability measured in terms of resistance that shall be a minimum of 2 megohms under the application for 1 minute, of an RMS voltage of 1000 volts, 60 Hertz, AC, plus twice the rated voltage of the applicable etcher when tested as specified in 4.4.1.

3.4.3 Plugs. Plugs on etcher power cables shall conform to UL 498 for two-pole, 3-wire grounding devices, rated at 15 amperes, 125 volts.

3.4.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means of preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or permanently deformed. Threaded parts shall be in the inch system and conform to Handbook H-28.

3.4.5 Type I and II. Type I and II etchers shall conform to UL 499 and each shall have components consisting of a case, transformer, control panel, power "on-off" switch, power cable with plug, etching tool lead, ground lead, etching tool or stylus, and a ground plate when furnished. The etchers shall withstand a 3-foot drop on any of the etcher case corners, on a concrete floor with no breakage or deformation so as to render the transformer unservicable when tested as specified in 4.4.2.

3.4.5.1 Heat output. The heat output of type I and II etchers shall be a minimum of 500 watts and 1000 watts respectively, with 4 heat-taps for type I and 10 for type II. The heat-taps shall vary in output from a minimum of 150 watts for each etcher to the maximum heat output of the applicable etcher when tested as specified in 4.4.3.

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3.4.5.2 Weight. The weight of type I and II etchers shall not exceed 25 and 35 pounds, respectively.

3.4.5.3 Case. The case of type I and II etchers shall be made of sheet steel of 20 gage (.0359 inch) minimum thickness. The base of the case shall have five embossed dimples of 3/8-inch spherical radius (inside) that shall raise the base of the etcher case. One dimple shall be located at the geometrical center and the remaining four dimples shall be located 3/4-inch from each side of each corner of the base. Alternatively, rubber pads of 7/8-inch diameter by 3/8-inch thick minimum dimensions may be substituted for the dimples. The pads shall be attached with locked screws. The case shall have a handle with a hand grip. Space shall be provided within the case for stowage of the power cable, etching tool with spare points, etching point and ground leads, and a ground plate when furnished. Alternatively, an integral part of the case, suitable for use as a ground for the etching tool, may be substituted for the ground plate (see 3.4.5.9).

3.4.5.4 Transformer. The transformer shall conform to UL 506 and shall be designed for the duty specified herein on a rated input voltage of 120 volts AC, without creating a fire hazard. The transformer shall be rigidly and securely mounted in the etcher case so that the transformer shall not become loosened or detached from its mounting. The use of an adhesive for mounting shall not be acceptable.

3.4.5.5 Control panel. The control panel shall have an indicator light to indicate when power is on, two output terminal studs with insulated knurled contact nuts, an "on" and "off" power switch, and a rotary heat-tap selector with a pointer and legible metallic dial for indicating the applicable heat-taps (see 3.4.5.1). The heat-tap dial shall be securely attached to the panel by rivets. Other means such as jack plug receptacles may be provided in lieu of the rotary heat-tap switch. The output terminals shall be insulated to prevent any grounding through the hole cross sectional and panel surface areas.

3.4.5.6 Power cable. The power cable for type I and II etchers shall be a minimum 5-foot length of commercial type SJT, SJO, SO or ST, three-conductor cable of suitable size. The cable shall be equipped with the plug specified in 3.4.3.

3.4.5.7 Etching tool and ground leads. Etching tool and ground leads shall be a minimum 3-foot length of suitable size cable conforming to asbestos-insulated wire, type AA of UL 115. Each lead shall be equipped with a spade type copper terminal at one end for connection to the transformer terminal studs. An etching tool shall be attached to the other end of the etching tool lead and a positive-type grounding copper terminal shall be attached to the other end of the ground lead.

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3.4.5.8 Etching tool. The etching tool or stylus shall have an insulated holder of suitable size and shape for ease of handling and a replaceable etching point. The tool shall be provided with a spare etching point. The etching point shall be copper, steel, or copper alloy and shall have integral tips or inserts of tungsten carbide or tungsten. Etching points in normal commercial use, of equal performance function as the points specified, shall be acceptable.

3.4.5.9 Ground plate. The ground plate of type I and II etchers, when furnished, shall have a minimum of 25 square inches of surface area. The plate shall be 18 gage (.0478 inch) steel minimum and copper plated for better electrical conductivity.

3.4.6 Type III. Type III etchers shall be similar to figure 3 and shall consist of a drive unit, reciprocating shaft, collet or chuck, switch, power cable, and replaceable etching points.

3.4.6.1 Drive unit. The drive unit shall consist of a heavy duty vibrating unit. The housing shall be sized and shaped to provide a comfortable hand-grip and shall have a power switch. The etcher shall operate to provide the specified speed for the reciprocating shaft (see 3.4.6.2).

3.4.6.2 Reciprocating shaft. The reciprocating shaft shall be mounted in a heavy duty sealed bearing. The shaft shall reciprocate at a minimum of 7200 strokes per minute when tested as specified in 4.4.4.

3.4.6.3 Chuck. A collet or chuck shall be located on the reciprocating shaft and shall accommodate replaceable etching points.

3.4.6.4 Switch. The "on-off" switch on the motor housing shall be placed in a convenient location for the operator. The switch shall be toggle, slide, or rocker-type, and shall be installed as part of the housing configuration. The switch shall conform to UL-45.

3.4.6.5 Etching points. The etching points for type III etchers shall be wear and shock resistant. The etcher shall be furnished with six alloy steel etching points consisting of two with diamond tips and four with tungsten carbide tips.

3.4.6.6 Power cable. The power cable for type III etcher shall be a minimum 5-foot length of commercial type SV, SVO, SVT, SJ, SJO or SJT, three conductor cable of suitable size. The cable shall be equipped with the plug specified in 3.4.3. The length of cable shall be measured from the point of entry into the tool.

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3.4.6.7 Weight. The weight of type III etcher shall not exceed 1 pound.

3.5 Fungus. When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed, need not be treated.

(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 0.001" to the component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.6 Performance.

3.6.1 Type I and II. Type I and II etchers, as applicable, shall be capable of operating for 2 hours at maximum heat output so that the transformer temperature (see 3.4.5.4) shall not exceed 65°C and the etching point shall withstand the heat developed without distortion or blunting of the tip so as to make etching illegible, when tested as specified in 4.4.5.

3.6.2 Type III etcher. Type III etchers shall be capable of operating for 1 hour so that the etching points shall withstand shock (see 3.4.6.5) without fracturing, bending or twisting, and the tip shall resist wear so as to make etching illegible when tested as specified in 4.4.5.

3.7 Finish. Etchers shall have enameled or plated finishes on metal surfaces in accordance with the manufacturer's commercial practice. Type I and II etchers shall have an enameled coat on the outside and inside surfaces of the case.

3.8 Identification plate. A corrosion resistant metal or plastic identification plate, permanently and legibly marked with the following information, shall be securely attached to the etcher.

Manufacturers name.
Manufacturer's model or serial number.
120 Volts.
50-60 Hertz, AC only.

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3.9 Instruction books. The manufacturer shall furnish two copies of his regular instruction book and parts list with each type I, II, and III etcher.

3.10 Accessories. Each etcher shall be furnished with such accessories as are normally furnished commercially.

3.11 Workmanship. Etchers shall be free of loose parts or components, burrs, sharp edges or slivers. The finish coat shall be smooth and free of runs, tackiness or areas of no finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specificatin where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with referenced specifications, drawings and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall be all etchers of one type offered for inspection at one time.

4.2.2.1 Visual inspection. The etchers shall be examined for the defects in table I. The inspection level shall be II with an AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Type	Not type specified	X	
Components	Part missing, not as specified fractured or broken	X	
	Insulation frayed or otherwise defective	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Identification plate	Not corrosion resistant metal or mounted as specified		X
	Not legible		X
Finish	Finish coat on type I and II not on inside and outside surfaces of case		X
Workmanship	Not free of loose parts or components, burrs, sharp edges or slivers		X
	Finish coat not smooth and free from runs, tackiness or areas of no finish		X

4.2.2.2 Dimensional examination. Etchers shall be examined for compliance with dimensions and weight specified. Any dimension or weight not within the specified tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. The first produced type I, II and III etcher, as applicable, shall be subjected to tests marked "X" in column 3 of table II. Failure to pass any test shall be cause for rejection of the initial production unit. Each sample etcher shall be tested as indicated in sequence 2 through 4 of table II. Failure to pass any test shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in defects per hundred units.

4.3 Code and standard examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

4.4 Test methods. The test schedule shall be in accordance with table II.

TABLE II. Test schedule

Test sequence	Quality conformance	First produced item	Test	Test paragraph	Requirement
1	2	3	4	5	6
1	Sample	X	Dielectric	4.4.1	3.4.2
2	Sample	X	Power output	4.4.3	3.4.5.1

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TABLE II. Test schedule (cont'd)

Test sequence	Quality conformance	First produced item	Test	Test paragraph	Requirement
1	2	3	4	5	6
3	Sample	X	Reciprocating shaft speed	4.4.4	3.5.2
4	-	X	Performance	4.4.5	3.6
5	-	X	Drop	4.4.2	3.4.5

4.4.1 Dielectric strength test. With the power switch turned "on", the insulation resistance of the etchers shall be measured to determine compliance with the requirements of 3.4.2 as the specified voltage is applied between the power input terminal of the applicable stretcher as the positive terminal and:

(1) Each of 2 ground terminals for type I and II etcher, as follows:

(a) A power output terminal with the etcher set on the maximum heat-tap, and
(b) the etcher case.

(2) The collet as the ground terminal for type III etcher. Any nonconformance shall constitute failure to pass this test.

4.4.2 Drop test. Type I and II etchers shall be drop tested to determine compliance with the no breakage and deformation requirements of 3.4.5. Any nonconformance shall constitute failure to pass this test.

4.4.3 Power output test. The power for type I and II etcher shall be turned "on" and the power at the output terminal for each heat-tap setting of the etcher measured to determine compliance with the heat output requirements of 3.4.5.1. Any nonconformance shall constitute failure to pass this test.

4.4.4 Reciprocating shaft speed test. Type III etcher shall be turned "on" for normal operation and the speed of the reciprocating shaft shall be measured to determine compliance with the speed requirements of 3.4.6.2. Any nonconformance shall constitute failure to pass this test.

4.4.5 Performance tests. Etchers shall be operated in accordance with the manufacturer's operating instructions. The applicable etcher shall be operated to etch letters and numerals on an 1/8-inch thick cold rolled steel plate of

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suitable area size. The etcher, in the time duration applicable (see 3.6.1 and 3.6.2), shall be operated continuously at 30-minute time periods with a 1-minute power cut-off and rest interval between periods. Any nonconformance with the applicable requirements of 3.6.1 and 3.6.2 shall constitute failure to pass this test.

4.5 Inspection of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing, and marking are in accordance with PPP-P-40 as required by section 5.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation, packaging, packing, and marking for each level of protection shall be in accordance with the applicable requirements of PPP-P-40 (see 6.2).

6. NOTES

6.1 Intended use.

6.1.1 Type I, light duty. Etchers, type I, are intended for etching tools and other objects of ferrous alloy material within the capacity of the tool.

6.1.2 Type II, heavy duty. Etchers type II are intended for use in etching heavy ferrous materials including tools and heavy castings, within the capacity of the tool.

6.1.3 Type III, etcher heavy duty. Type III etcher is intended for use in etching practically all materials including ferrous alloy materials.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Level of preservation and packaging and level of packing required (see 5.1).

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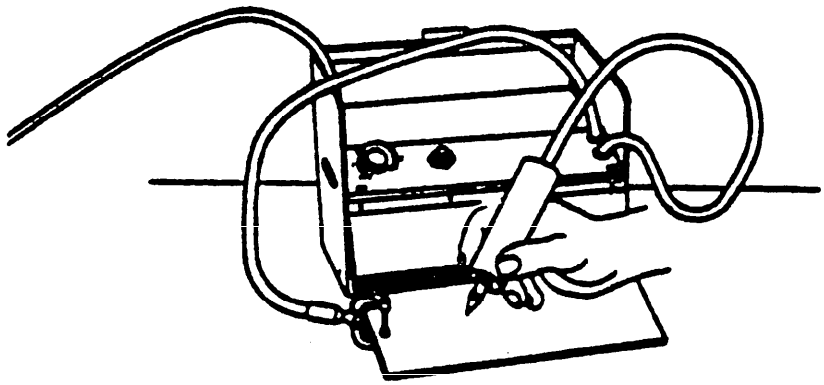


FIGURE 1. *Type I, etcher — light duty.*

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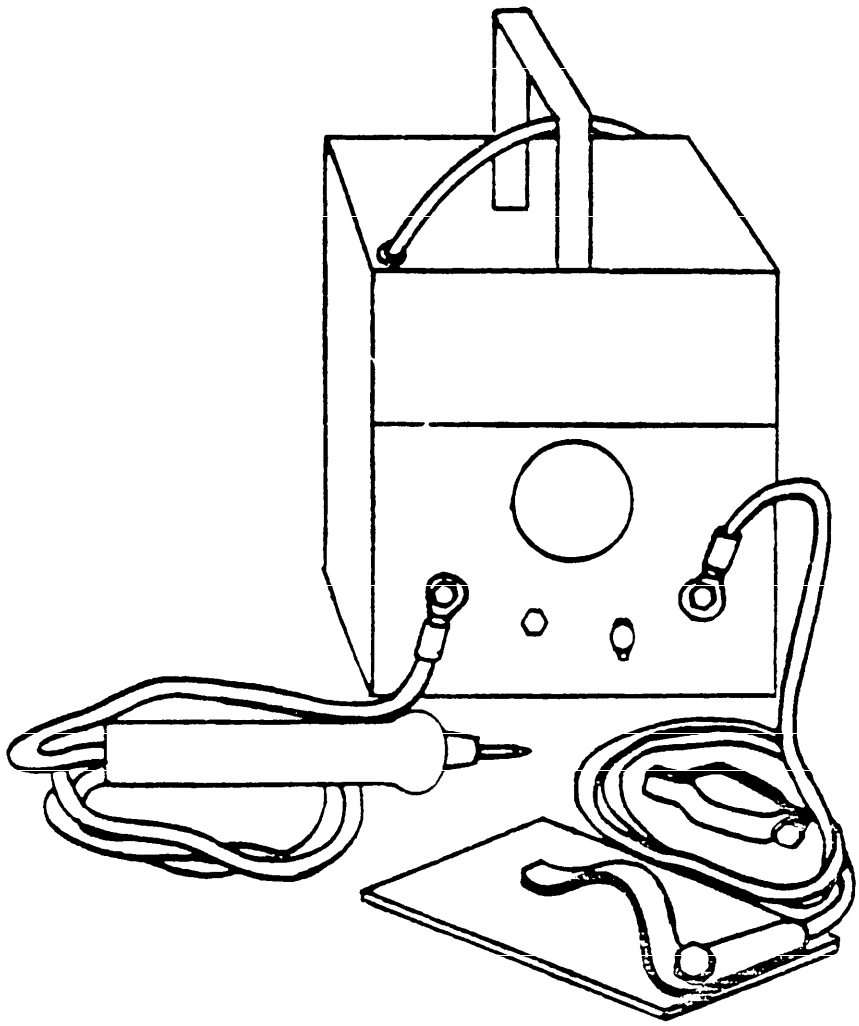


FIGURE 2. TYPE II ETCHER —HEAVY DUTY

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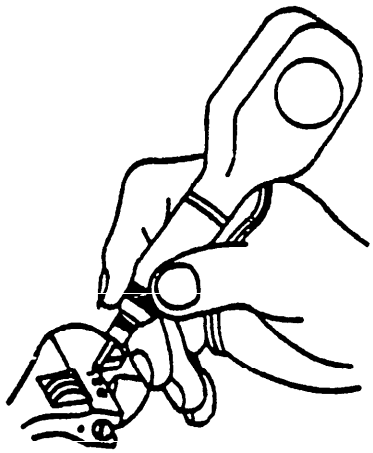


FIGURE 8. *Type III, etcher — heavy duty vibrating type.*

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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